

Statistical Inference

Technical reference

Companion to the note on lorenzoliuzzo.github.io. Statements, conditions and proofs. Probability background is in the Probability Foundations reference.

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This chapter uses the probability background (axioms, expectation and variance rules, limit theorems, Slutsky) from the Probability Foundations reference and the sampling distributions of the mean and variance, the χ^2 , t and F laws, from the Sampling Distributions reference. Both are quoted without proof.

1 Point estimation

Let $X_1, \dots, X_n \sim f(\cdot; \theta)$, $\theta \in \Theta \subset \mathbb{R}$, and $\hat{\theta} = T(X_1, \dots, X_n)$.

Definition. $\text{bias}(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta$. The estimator is unbiased if this vanishes for all θ . It is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$. Its mean squared error is $\text{MSE}(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$.

Theorem (Bias–variance decomposition).

$$\text{MSE}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}(\hat{\theta})^2. \quad (1)$$

Proof. Add and subtract $\mathbb{E} \hat{\theta}$: $(\hat{\theta} - \theta)^2 = (\hat{\theta} - \mathbb{E} \hat{\theta})^2 + 2(\hat{\theta} - \mathbb{E} \hat{\theta})(\mathbb{E} \hat{\theta} - \theta) + (\mathbb{E} \hat{\theta} - \theta)^2$. The cross term has zero expectation because $\mathbb{E} \hat{\theta} - \theta$ is a constant and $\mathbb{E}[\hat{\theta} - \mathbb{E} \hat{\theta}] = 0$. $\square$

Example (A biased estimator with smaller MSE). For $X_i \sim \mathcal{N}(\mu, \sigma^2)$, the estimator $\hat{\sigma}_c^2 = c \sum_i (X_i - \bar{X})^2$ has, using $\sum_i (X_i - \bar{X})^2 \sim \sigma^2 \chi_{n-1}^2$ (mean $(n-1)\sigma^2$, variance $2(n-1)\sigma^4$),

$$\text{MSE} = \sigma^4[2c^2(n-1) + ((n-1)c - 1)^2]. \quad (2)$$

This is minimized at $c = 1/(n+1)$, not at the unbiased choice $c = 1/(n-1)$. The unbiased estimator is therefore not MSE-optimal, a first instance of the shrinkage idea behind Ridge and Lasso.

1.1 Fisher information and the Cramér–Rao bound

Assume the regularity conditions: the support of $f(\cdot; \theta)$ does not depend on θ , $\log f$ is twice differentiable in θ , and differentiation under the integral sign is allowed.

Definition (Score and Fisher information). The score of one observation is $s(x; \theta) = \partial_\theta \log f(x; \theta)$, and

$$I(\theta) = \mathbb{E}_\theta[s(X; \theta)^2] = \text{Var}_\theta(s(X; \theta)). \quad (3)$$

Lemma. Under the regularity conditions, $\mathbb{E}_\theta[s(X; \theta)] = 0$ and $I(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log f(X; \theta)]$. For n independent observations the information is $nI(\theta)$.

Proof. $\mathbb{E}[s] = \int (\partial_\theta f / f) f \, dx = \partial_\theta \int f \, dx = \partial_\theta 1 = 0$. Differentiating $\int s f \, dx = 0$ once more gives $\int (\partial_\theta s) f \, dx + \int s^2 f \, dx = 0$, and $\partial_\theta s = \partial_\theta^2 \log f$. Additivity follows because the log-likelihood of independent data is the sum of the individual ones, so the scores add and, being independent with mean zero, their variances add. $\square$

Theorem (Cramér–Rao lower bound). If $\hat{\theta}$ is unbiased with finite variance, then $\text{Var}_\theta(\hat{\theta}) \geq 1/(nI(\theta))$.

Proof. Let $U = \sum_i s(X_i; \theta)$ be the total score, so $\mathbb{E}U = 0$ and $\text{Var}(U) = nI(\theta)$. Unbiasedness reads $\int \hat{\theta}(x) f_{n(x; \theta)} \, dx = \theta$ (f_n the joint density). Differentiating in θ gives $\int \hat{\theta}(x) U(x) f_n \, dx = 1$, i.e. $\text{Cov}(\hat{\theta}, U) = \mathbb{E}[\hat{\theta}U] = 1$ since $\mathbb{E}U = 0$. Cauchy–Schwarz: $1 = \text{Cov}(\hat{\theta}, U)^2 \leq \text{Var}(\hat{\theta}) \text{Var}(U) = \text{Var}(\hat{\theta}) nI(\theta)$. $\square$

Remark. For a biased estimator with $b(\theta) = \text{bias}(\hat{\theta})$ the same argument gives $\text{Var}(\hat{\theta}) \geq (1 + b'(\theta))^2 / (nI(\theta))$. Equality in Cauchy–Schwarz holds iff $\hat{\theta} - \theta$ is proportional to the score, which happens exactly in exponential families for the mean-value parameter.

1.2 Sufficiency (brief)

A statistic $T(X)$ is sufficient for θ if the conditional law of the sample given T does not depend on θ . By the Fisher–Neyman factorization theorem, T is sufficient iff $f_{n(x; \theta)} = g(T(x); \theta)h(x)$. For normal data with known σ , $\sum_i X_i$ is sufficient for μ ; for the Bernoulli model, $\sum_i X_i$ is sufficient for p . The Rao–Blackwell theorem states that conditioning an estimator on a sufficient statistic never increases its MSE.

2 Maximum likelihood

Definition. The likelihood is $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$, the log-likelihood $\ell(\theta) = \log L(\theta)$ and the maximum likelihood estimator $\hat{\theta}_{\text{MLE}} = \arg \max_\theta \ell(\theta)$.

Example (Bernoulli). $\ell(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1 - p)$, $\ell'(p) = (\sum x_i)/p - (n - \sum x_i)/(1 - p)$. Setting it to zero gives $\hat{p} = \bar{x}$; $\ell'' < 0$ so it is a maximum. The information is $I(p) = 1/(p(1 - p))$ and $\text{Var}(\hat{p}) = p(1 - p)/n = 1/(nI(p))$: the MLE attains the Cramér–Rao bound exactly.

Example (Normal). $\ell(\mu, \sigma^2) = -(n/2) \log(2\pi\sigma^2) - (2\sigma^2)^{-1} \sum_i (x_i - \mu)^2$. The first-order conditions give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = n^{-1} \sum_i (x_i - \bar{x})^2$. By the theorem on S^2 , $\mathbb{E} \hat{\sigma}^2 = ((n - 1)/n) \sigma^2$: the MLE of the variance is biased, but consistent, and $\hat{\sigma}^2 = c S^2$ with $c = (n - 1)/n \rightarrow 1$.

2.1 Asymptotic properties

Theorem (Asymptotics of the MLE). Under the regularity conditions above, with θ_0 the true parameter and $I(\theta_0) > 0$: (i) $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0$; (ii) $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$; (iii) invariance: the MLE of $g(\theta)$ is $g(\hat{\theta}_{\text{MLE}})$, and by the delta method it is asymptotically $\mathcal{N}(g(\theta_0), g'(\theta_0)^2/(nI(\theta_0)))$.

Proof. Sketch of (ii). Expand the score at the MLE around θ_0 : $0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + (\hat{\theta} - \theta_0)\ell''(\tilde{\theta})$ for some $\tilde{\theta}$ between them. Hence

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{n^{-1/2}\ell'(\theta_0)}{-n^{-1}\ell''(\tilde{\theta})}. \quad (4)$$

The numerator is $n^{-1/2} \sum_i s(X_i; \theta_0)$, a normalized sum of i.i.d. mean-zero variables with variance $I(\theta_0)$, so by the CLT it tends to $\mathcal{N}(0, I(\theta_0))$. By the law of large numbers and consistency, the denominator tends to $-E[\partial_{\theta}^2 \log f] = I(\theta_0)$. Slutsky gives the limit $\mathcal{N}(0, I(\theta_0)^{-1})$. $\square$

The practical consequence is the standard error $\hat{\text{se}} = 1/\sqrt{nI(\hat{\theta})}$ (or its observed-information version, $1/\sqrt{-\ell''(\hat{\theta})}$), valid for large n .

2.2 Method of moments

Equate the first d population moments $m_{j(\theta)} = E_{\theta} X^j$ to the sample moments $n^{-1} \sum_i X_i^j$ and solve for $\theta \in \mathbb{R}^d$. The estimator is consistent when $m = (m_1, \dots, m_d)$ is invertible with continuous inverse, but generally has larger asymptotic variance than the MLE.

3 Interval estimation

Definition. A $(1 - \alpha)$ confidence interval is a random interval $[L(X), U(X)]$ with $P_{\theta}(L(X) \leq \theta \leq U(X)) \geq 1 - \alpha$ for all $\theta \in \Theta$. The left side is the coverage; the inequality is replaced by an equality for exact intervals.

The probability refers to the random endpoints under repeated sampling. For the realized interval, θ is fixed and the event either holds or not.

Definition (Pivot). $Q(X; \theta)$ is a pivot if its distribution does not depend on θ .

If $P(a \leq Q(X; \theta) \leq b) = 1 - \alpha$ and Q is monotone in θ , solving $a \leq Q \leq b$ for θ gives an exact interval.

Theorem (Intervals for the normal model). Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d.. Exact $(1 - \alpha)$ intervals:

- mean, σ unknown: $\bar{X} \pm t_{n-1, 1-\alpha/2} S / \sqrt{n}$;
- mean, σ known: $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$;
 $[(n-1)S^2 / \chi_{n-1, 1-\alpha/2}^2, (n-1)S^2 / \chi_{n-1, \alpha/2}^2]$
- variance: $(n-1)S^2 / \chi_{n-1, \alpha/2}^2$

Proof. $P(-t \leq T \leq t) = 1 - \alpha$ with $t = t_{n-1, 1-\alpha/2}$ and T the pivot of the sampling-distribution theorem; rearranging $-t \leq (\bar{X} - \mu)/(S/\sqrt{n}) \leq t$ gives the first interval. The second is identical with T replaced by $\sqrt{n}(\bar{X} - \mu)/\sigma \sim \mathcal{N}(0, 1)$. For the third, $P(\chi_{n-1, \alpha/2}^2 \leq (n-1)S^2/\sigma^2 \leq \chi_{n-1, 1-\alpha/2}^2) = 1 - \alpha$ and inverting in σ^2 reverses the inequalities. $\square$

Theorem (Wald interval). If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta))$ with v continuous, then $\hat{\theta} \pm z_{1-\alpha/2} \sqrt{v(\hat{\theta})/n}$ has asymptotic coverage $1 - \alpha$. For the MLE, $v = I^{-1}$. For a proportion, $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$.

Proof. $\sqrt{n}(\hat{\theta} - \theta)/\sqrt{v(\hat{\theta})} = [\sqrt{n}(\hat{\theta} - \theta)/\sqrt{v(\theta)}] \cdot [\sqrt{v(\theta)/v(\hat{\theta})}]$; the first factor tends in distribution to $\mathcal{N}(0, 1)$ and the second in probability to 1 by consistency and continuity. Apply Slutsky. $\square$

Remark. The Wald interval for a proportion has coverage that oscillates and can be far below $1 - \alpha$ when n is small or p is near 0 or 1. The Wilson interval, obtained by inverting the score test instead of replacing p by $\hat{p}$ in the variance, is the standard fix.

4 Hypothesis testing

4.1 Framework

A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is a rule $\varphi(X) \in \{0, 1\}$ (1 means reject), typically $\varphi = \mathbb{1}\{T(X) \in R\}$ for a statistic T and rejection region R .

Definition. The size of the test is $\sup_{\theta \in \Theta_0} P_\theta(\varphi = 1)$; it has level α if the size is at most α . The power function is $\beta(\theta) = P_\theta(\varphi = 1)$, so the Type II error at $\theta \in \Theta_1$ is $1 - \beta(\theta)$.

Definition. The p -value of observed t_{obs} for a statistic T that rejects for large values is $p = \sup_{\theta \in \Theta_0} P_\theta(T \geq t_{\text{obs}})$.

Lemma (Uniformity of the p -value). If H_0 is simple, T has a continuous distribution function G under H_0 , and rejection is for large T , then $p(T) = 1 - G(T) \sim \text{Uniform}(0, 1)$ under H_0 . Hence rejecting when $p \leq \alpha$ has size exactly α .

Proof. $G(T) \sim \text{Uniform}(0, 1)$ by the probability integral transform: $P(G(T) \leq u) = P(T \leq G^{-1}(u)) = u$. Then $1 - G(T)$ is also uniform. $\square$

4.2 The t -tests

Theorem (One-sample and Welch t -tests). (i) For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d., under $H_0 : \mu = \mu_0$ the statistic $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$ has the t_{n-1} distribution, so rejecting for $|T| > t_{n-1, 1-\alpha/2}$ has size α . (ii) For two independent normal samples with means μ_1, μ_2 , and $H_0 : \mu_1 = \mu_2$,

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad (5)$$

is approximately t_ν with

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1 - 1) + (S_2^2/n_2)^2/(n_2 - 1)}. \quad (6)$$

Part (i) is the pivot theorem with $\mu = \mu_0$. In (ii) the denominator is a weighted sum of independent scaled χ^2 variables, which is not exactly a scaled χ^2 ; the Welch-Satterthwaite ν matches the first two moments of a scaled χ_ν^2 to it. With equal variances the pooled test is exactly $t_{n_1+n_2-2}$, but Welch is robust to unequal variances at little cost.

Theorem (Duality of tests and intervals). If, for each θ_0 , $A(\theta_0)$ is the acceptance region of a level- α test of $\theta = \theta_0$, then $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a $(1 - \alpha)$ confidence set. Conversely, from a confidence set one obtains a test by rejecting $\theta = \theta_0$ iff $\theta_0 \notin C(x)$.

Proof. $P_{\theta_0}(\theta_0 \in C(X)) = P_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha$, for every θ_0 . The converse is the same identity read backwards. $\square$

4.3 Power and sample size

Theorem (Power of the one-sided z -test). Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with σ known, and reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu > \mu_0$ when $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma > z_{1-\alpha}$. Then

$$\beta(\mu_1) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma), \quad (7)$$

and $\beta(\mu_1) \geq 1 - \beta_0$ holds iff $n \geq \left((z_{1-\alpha} + z_{1-\beta_0})\sigma/(\mu_1 - \mu_0) \right)^2$.

Proof. Under $\mu = \mu_1$, $\sqrt{n}(\bar{X} - \mu_1)/\sigma \sim \mathcal{N}(0, 1)$, so $Z = Z_0 + \sqrt{n}(\mu_1 - \mu_0)/\sigma$ with $Z_0 \sim \mathcal{N}(0, 1)$. Hence $P(Z > z_{1-\alpha}) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma)$. Requiring this to be at least $1 - \beta_0$ means $z_{1-\alpha} - \sqrt{n}\delta/\sigma \leq -z_{1-\beta_0}$ (with $\delta = \mu_1 - \mu_0$), and solving for n gives the bound. $\square$

The required n scales like σ^2/δ^2 : halving the detectable effect quadruples the sample.

4.4 Likelihood ratio tests

Theorem (Neyman–Pearson lemma). For simple $H_0 : \theta = \theta_0$ against simple $H_1 : \theta = \theta_1$, let φ^* reject when $L(\theta_1)/L(\theta_0) > k$, with k chosen so that φ^* has size α . Then for any other test φ of size at most α , $\beta_\varphi(\theta_1) \leq \beta_{\varphi^*}(\theta_1)$.

Proof. For every x , $(\varphi^*(x) - \varphi(x))(f(x; \theta_1) - kf(x; \theta_0)) \geq 0$: where $\varphi^* = 1$ the second factor is positive, where $\varphi^* = 0$ it is non-positive and the first factor is $-\varphi(x) \leq 0$ or zero. Integrating over x gives $\beta_{\varphi^*}(\theta_1) - \beta_\varphi(\theta_1) \geq k(\alpha_{\varphi^*} - \alpha_\varphi) \geq 0$, since both sizes are at most α and φ^* has size exactly α . $\square$

Theorem (Wilks' theorem). Let $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$, where Θ has d free parameters and Θ_0 has $d - r$. Under the regularity conditions and H_0 ,

$$-2 \log \Lambda \xrightarrow{d} \chi_r^2. \quad (8)$$

The proof expands ℓ quadratically around the MLE and uses asymptotic normality of the MLE, so that $-2 \log \Lambda$ becomes a quadratic form in r asymptotically independent standard normals. The Wald and score tests are asymptotically equivalent to it. In the linear model with Gaussian errors, the likelihood ratio is a monotone function of an F statistic, which is why the F -test of regression has an exact distribution.

4.5 Multiple testing

With m tests and all nulls true, each at level α , the expected number of false rejections is $m\alpha$.

Theorem (Bonferroni and Benjamini–Hochberg). (i) Rejecting each hypothesis with $p_i \leq \alpha/m$ controls the family-wise error rate, $P(\text{at least one false rejection}) \leq \alpha$, with no assumption on the dependence of the p -values. (ii) Order the p -values $p_{(1)} \leq \dots \leq p_{(m)}$ and reject the k smallest, where $k = \max\{i : p_{(i)} \leq i\alpha/m\}$. If the p -values are independent, the false discovery rate $E[V/\max(R, 1)]$ (with V false rejections out of R) is at most $\alpha m_0/m \leq \alpha$, where m_0 is the number of true nulls.

Proof. (i) By the union bound, $P\left(\bigcup_{i \in H_0} \{p_i \leq \alpha/m\}\right) \leq m_0\alpha/m \leq \alpha$. (ii) is the Benjamini–Hochberg (1995) theorem; the proof conditions on the other p -values and is omitted. $\square$

5 Summary of the main formulas

Quantity	Result
Standard error of the mean	$\sigma/\sqrt{n}$
S^2 unbiased	$E S^2 = \sigma^2$, divisor $n - 1$

Normal sampling laws	$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2, T \sim t_{n-1}, \bar{X} \perp S^2$
MSE	$\text{Var} + \text{bias}^2$
Cramér–Rao	$\text{Var}(\hat{\theta}) \geq 1/(nI(\theta))$
MLE	$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$
CI for the mean	$\bar{X} \pm t_{n-1, 1-\alpha/2} S/\sqrt{n}$
CI for the variance	$[(n-1)S^2/\chi_{n-1, 1-\alpha/2}^2, (n-1)S^2/\chi_{n-1, \alpha/2}^2]$
Wald interval	$\hat{\theta} \pm z_{1-\alpha/2} \hat{\text{se}}$
Sample size, one-sided z -test	$n = ((z_{1-\alpha} + z_{1-\beta})\sigma/\delta)^2$
Likelihood ratio	$-2 \log \Lambda \xrightarrow{d} \chi_r^2$
Bonferroni / BH	$p_i \leq \alpha/m$ / largest k with $p_{(k)} \leq k\alpha/m$

6 References

- G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed., Duxbury, 2002: the standard source for the proofs of the Cramér–Rao bound, the MLE asymptotics and the Neyman–Pearson lemma.
- J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, Cengage, 2011: the course text for the first part of the syllabus.
- L. Wasserman, *All of Statistics*, Springer, 2004: concise statements of the limit theorems, MLE and multiple testing.
- Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate”, *J. R. Statist. Soc. B* 57 (1995).